AI AGENT BUILDER INTERN · CASE STUDY FOR HEIDI SYSTEMS

Finding Berlin's property managers for Heidi

How I built an AI agent pipeline that turns public sources into a ranked, reachable lead list, with proof for every fact.

1,115
qualified leads

456
direct first-person contacts

≈ $0.09
per lead

Apsal Kalathukunnel · November 2026

THE PROBLEM

Good leads are scattered, hidden and time-sensitive

No single list

Property managers are spread over maps, association directories and thousands of small websites.

Thin data where they are listed

Directories give a name and a phone; size, billing setup and meters sit deep in company pages, if anywhere.

The buyer is hidden

Websites show info@ and a switchboard, not the person who decides.

Timing decides

Needs change with deadlines, contracts and growth; a list goes stale fast.

WHAT SALES NEEDS

- Who to call: the person who runs the buildings
- Why they fit, and why now
- How to reach them: phone, email, LinkedIn
- Proof behind every claim

RESULT AT A GLANCE

2,093 companies in, 1,115 ranked leads out

2,093

companies found

maps, directories, job ads

1,115

qualified leads

A 49 · B 849 · C 217

731

named decision-makers

66% of leads, instead of info@

456

direct first-person contacts

a phone or e-mail next to the name

~6,000

facts with a source quote

every value traceable to a page

≈ $0.09

per lead

model cost for the whole run

KPI 1 · FIT SCORE (0–100)

Is this a good customer for Heidi?

Manages apartments	+25	Heidi bills heating costs for residential buildings: the core customer.
Meters read by hand	+25	Heating Cost Ordinance (HKVO): meters must be remote-readable from 2027, an urgent, dated need.
Provider change announced	+25	Contract end, planned switch or tender: an open buying window.
Portfolio size	+15	Revenue scales with apartments; 1,000+ = key account (+15), 100+ = +10.
Current billing known	+10	In-house billers or a named competitor: we know what to replace.

A · call first · fit 50+ · 49

B · good prospect · fit 25–49 · 849

C · weaker fit · fit 1–24 · 217

KPI 2 · ACCESS SCORE (0–100)

Can we reach the person who runs the buildings?

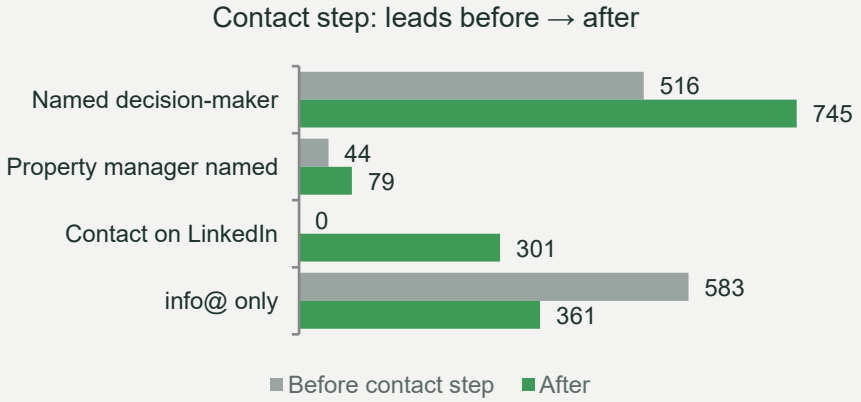
The ladder

Property manager, direct line	100
Property manager, name only	80
Owner / director, direct line	60
Owner / director, name only	40
Other staff	30
info@ or switchboard only	10

+10 for a personal LinkedIn profile (max 100)

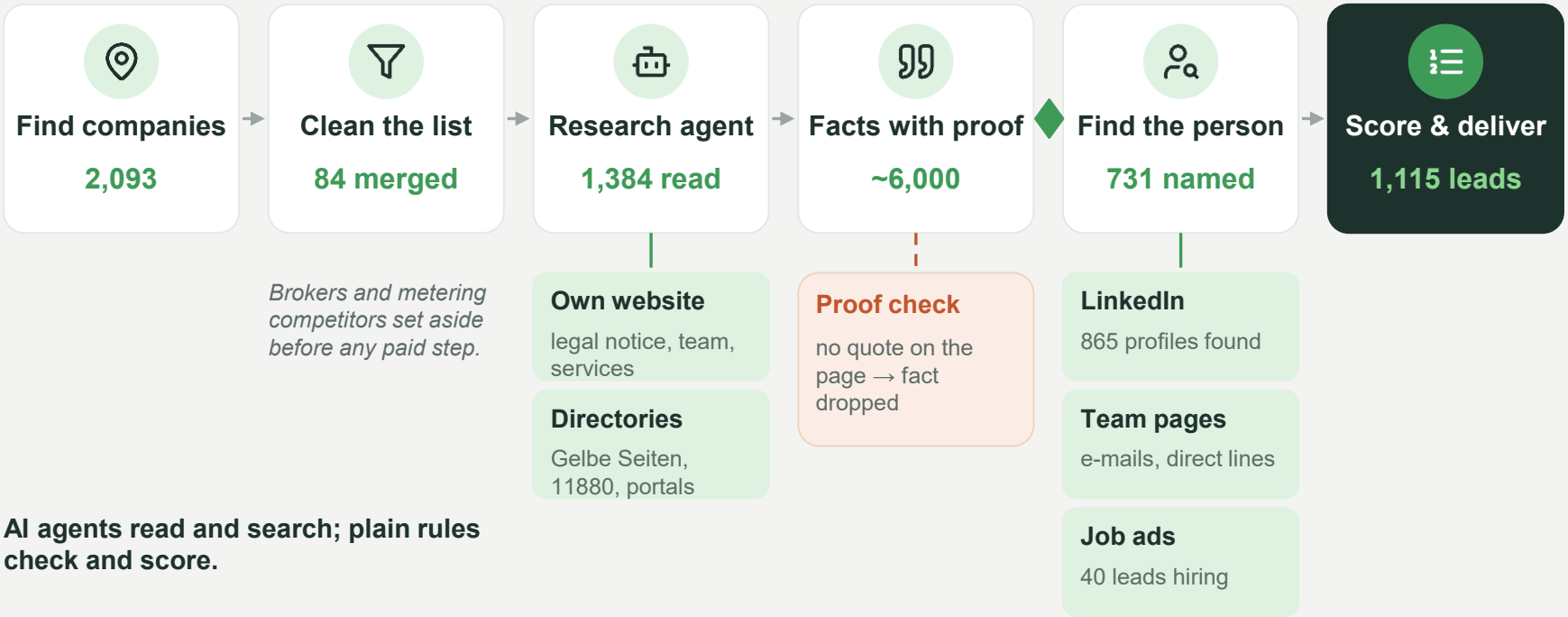
Why a separate score

A great customer we can't reach is not a first call. The property manager who runs the buildings can push adoption upward, even inside a big firm.



HOW IT WORKS

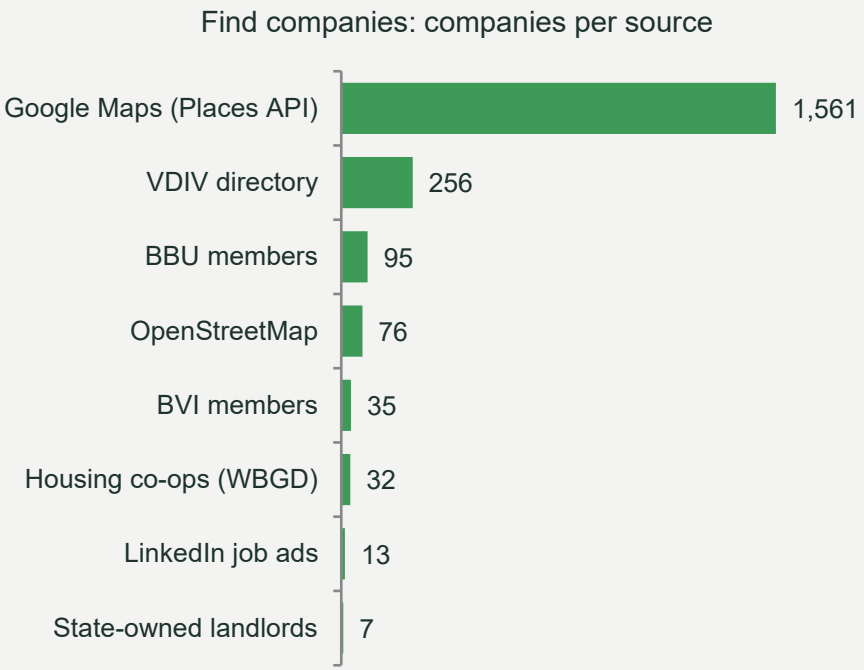
Six stages, with agents where judgement is needed



AI agents read and search; plain rules check and score.

SOURCES

Public sources, each chosen for what sales needs



FACTS

Company websites

1,384 read: legal notice, team, contact and service pages give every fact and most contacts. Directory and listing pages only for firms without a site.

PEOPLE

LinkedIn, team pages, job ads

865 LinkedIn profiles; personal e-mails and direct lines from team pages; 77 job ads (40 leads hiring).

WHY THESE

Value for sales

Maps give address, phone and map position; VDIV membership is a warm intro (Heidi is VDIV's partner).

TOOLS AND AUTOMATIONS

The toolchain: one command, five jobs

1 Discover	**Google Places API** 1,561 offices, phone, location	**OpenStreetMap** 76 offices from open map data	**Member directories** VDIV, BBU, BVI and co-op lists
2 Collect	**Scrapy crawler** Own pages, saved as snapshots	**Playwright** Reads JavaScript-only websites	
3 Understand	**Claude Agent SDK** Haiku: pages → facts with quotes	**Claude WebSearch** Missing websites, LinkedIn profiles	**LinkedIn job search** 647 public ads: hiring and contacts
4 Store & check	**SQLite evidence store** Every value links back to its page	**Pydantic + rules** Quotes and numbers must match	
5 Deliver	**Sales dashboard** Map, filters, call card	**Excel workbook** Sales list, evidence, definitions	

One command runs all five jobs, built on the Claude Agent SDK · 150+ automated tests guard the checks

HOW THE DATA WAS VALIDATED

Three layers between a web page and the sales list

1 · Text matching

- The quote must be on the saved page (exact, or ≥ 80% of its words)
- Numbers, phones and e-mails must appear in the quote
- LinkedIn people only if link and name came back from the search

2 · Data types and rules

- Strict schema: field, category, number, scope, date
- Fixed categories (WEG, rental, own stock …)
- Ranges ignored; company totals never counted as Berlin; old facts never current

3 · AI checks

- The model must quote and categorise every fact
- Rejected facts go back once, with the reason
- A second pass looks for missing facts; current-job check for contacts

Plus: Berlin required, duplicates merged, brokers, competitors and service firms set aside, with hand checks of samples. A separate AI reviewer ran in the pilots and was dropped for speed and cost; the code checks stayed.

VALUE GENERATED

What the list gives Heidi's sales team

115
key accounts
1,000+ apartments

333k
apartments in Berlin
published counts only

25
meters read by hand
HKVO upgrade due

40
hiring property staff
growing, short-handed

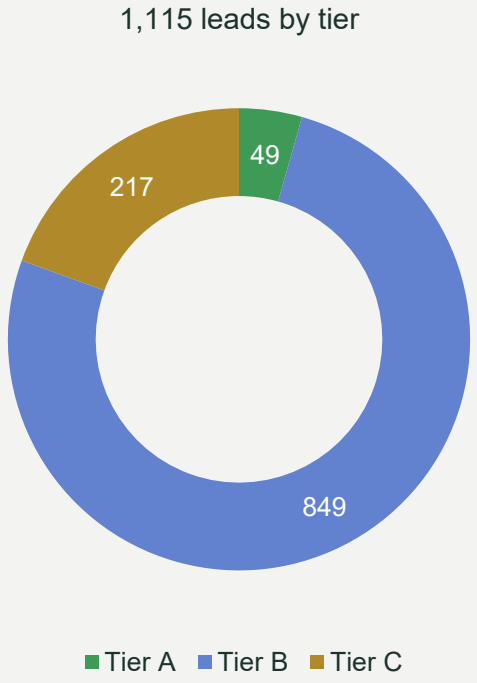
227
VDIV members
warm intro via Heidi's partner

149
billing setup known
in-house or named provider

≈ 230–350 hours

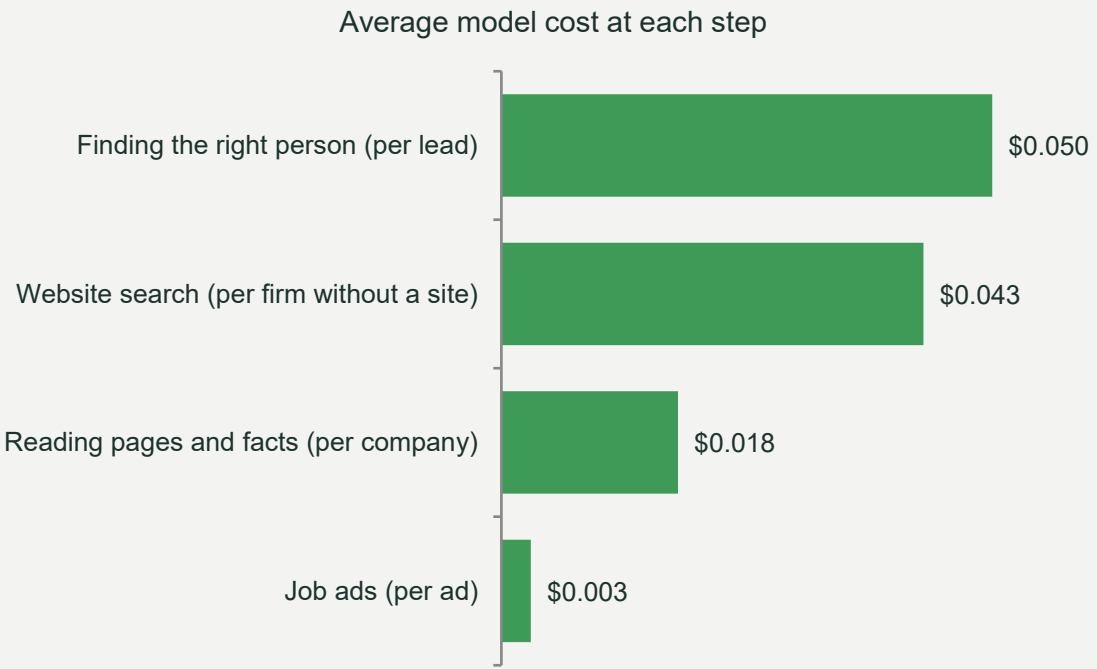
Projection: 1,384 companies at 10–15 min each by hand.

of manual research replaced, at ≈ $0.09 per lead



COST

Under ten cents per lead



≈ $0.09

per lead overall (1,115 leads)

Paid steps run only on companies still in the list; brokers and competitors are removed first.

KPI 3 · IS THE PIPELINE WORKING?

Four numbers to watch on every run

41%

direct first-person contact rate

456 of 1,115 leads

Reachability is the bottleneck for sales; this is the number to push up.

100%

facts with a source quote

~6,000 facts kept

Trust: a salesperson can open the page behind every value.

≈ $0.09

cost per lead

model usage, whole run

Shows the pipeline scales to all of Germany at a small cost.

66%

named decision-maker rate

731 of 1,115 leads

Sales calls a person, not a switchboard.

DESIGN DECISIONS

Choices that shaped the result

Proof for every fact

Sales can trust and quote what they see; nothing is invented.

Fit and reach kept apart

A perfect customer without a contact isn't a first call.

AI reads, rules score

Cheap, explainable scores; the model only reads and searches.

Company sources first

Own pages first; an exact published number beats an approximate one, which is labelled.

No logins, private contacts

Public pages only; names stay out of git and shared pages.

Utility over proof

Trust the company's own pages; check numbers, not wording.

LIMITS AND NEXT STEPS

Honest gaps, then the natural next steps

Limits today

- Personal e-mails are rarely public
- LinkedIn results often lack job titles
- Metering provider and contract dates rarely published (1–2%)

More value per lead

- Job titles from a licensed LinkedIn data provider
- Verify likely e-mails before outreach
- Weekly re-checks for hiring and provider changes
- Learn from CRM outcomes to tune the fit score

Scale

- All of Germany: map search by city and postcode, regional directories
- Re-read only pages that changed
- Scheduled monthly runs

WHAT I DELIVERED

A pipeline, a sales tool and the proof behind it

Sales dashboard

Berlin map, filters, call card with direct lines, LinkedIn and the evidence behind every value.

Excel workbook

Companies (A–C), Details, Evidence (7,216 quotes) and Definitions, CRM columns ready.

How-it-works page

Animated pipeline, sources, checks and costs for anyone new to the list.

Tested, re-runnable code

Python pipeline on the Claude Agent SDK; 150+ automated tests; contact data kept out of git.

WHAT THIS SHOWS

How I would build agents at Heidi

Start from the sales outcome

KPIs in Heidi's own terms: customer fit, reachability, the Heating Cost Ordinance upgrade.

Agents where judgement is needed

AI reads and searches; plain rules check and score. Explainable, at ≈ $0.09 per lead.

Trust built in

Proof behind every fact, hand checks of samples, limits stated openly.

Ship tools people use

A dashboard, a workbook and a how-it-works page a salesperson can use on day one.

Thank you · Apsal Kalathukunnel